

# LITHIUM BATTERY FIRES: 7 RULES AND THE WARNING SIGNS

No extinguisher you own stops thermal runaway. The whole game is making sure it never starts.

A lithium ion cell in thermal runaway makes its own oxygen inside the sealed can, so a powder extinguisher has nothing to smother. In a federal lab test, a dry chemical extinguisher buried the pack in powder for about 45 seconds. Then it reignited and exploded. Only continuous cooling works, and that means the fire department's hose, not your bucket.

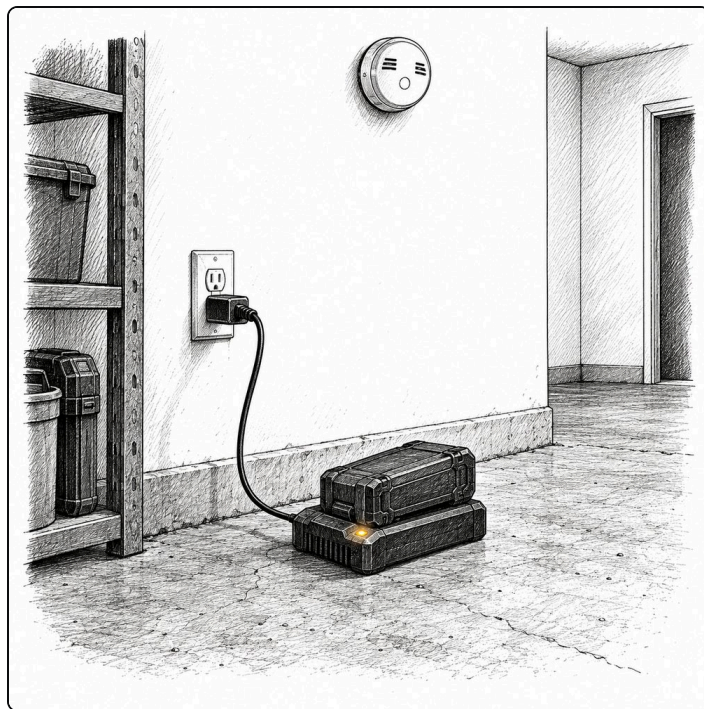
## THE 7 RULES

- 1 **Charge on the floor**, on a hard surface. Never on a bed, a couch, or under a pillow.
- 2 **Never charge across your way out**. Not by the front door, in the hall, by the bedroom door, or under a window you would climb through.
- 3 **Unplug when it is full**, and never charge while you sleep.
- 4 **Wall outlet only**. No power strip, no extension cord.
- 5 **Only the charger that came with it, and only certified packs**. Look for the UL 2849 mark on e bikes, UL 2271 on the battery.
- 6 **Room temperature**. Do not charge below 32 F or above 105 F. Let a cold pack warm up first.
- 7 **Inspect it, and learn the warning signs in order**.

## THE WARNING SIGNS, IN ORDER

- ☐ Too hot to hold comfortably
- ☐ Swelling or bulging
- ☐ A sweet chemical smell
- ☐ Hissing, cracking, or popping
- ☐ Smoke

**At the first sign:** stop using it, unplug it, and if it is safe, move it away from anything that burns with tongs or gloves. Then leave and call. **At smoke:** everyone out, close the door behind you, call 911.



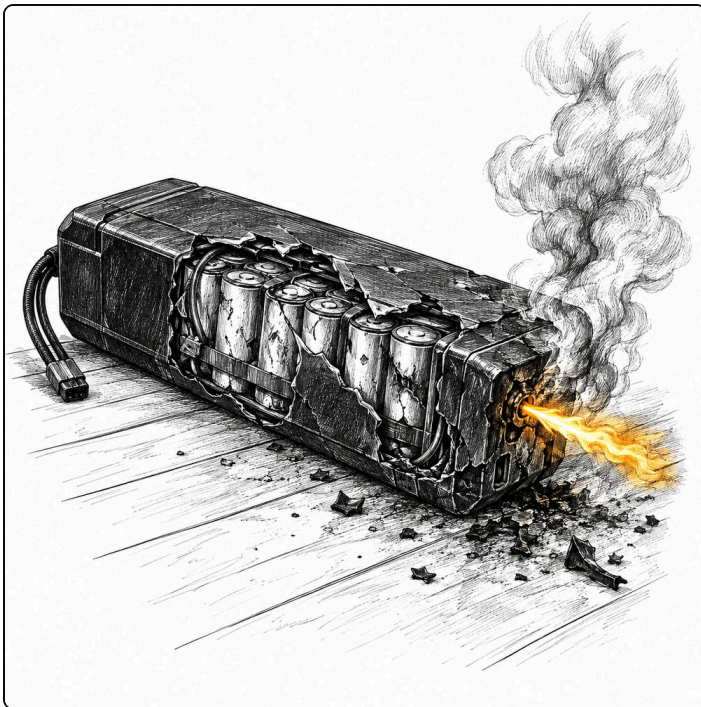
On the floor, wall outlet, away from the exit, smoke alarm overhead.

## WHY SPEED MATTERS

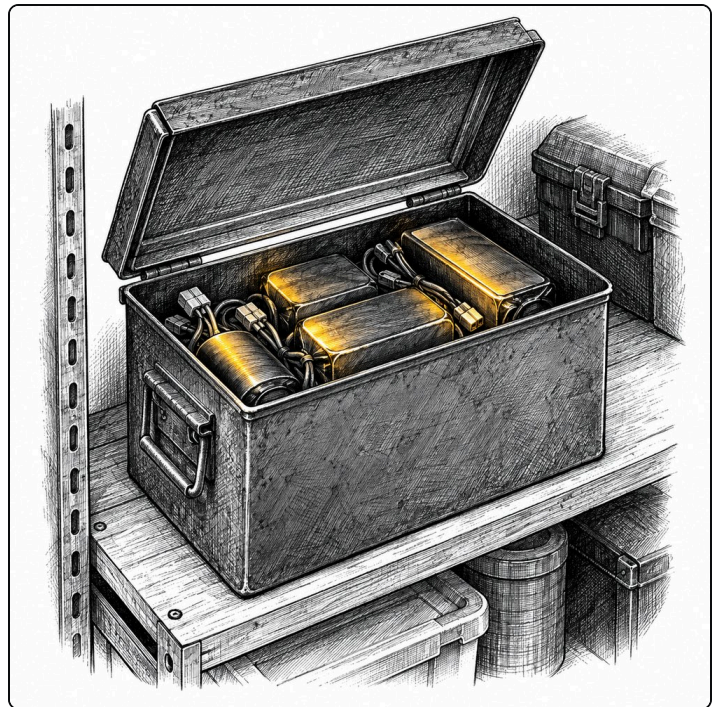
|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| <b>12 sec</b>  | First visible smoke to the bedroom window blowing out, in the e scooter burn tests.              |
| <b>40 sec</b>  | Flames coming out the window. Flashover came within 30 seconds of the window failing.            |
| <b>13 sec</b>  | For the scooter to hit the peak heat of an overstuffed chair fire. A chair takes 2 or 3 minutes. |
| <b>18 to 1</b> | New York City battery fire deaths, 2023 to 2025, after a law required certified packs.           |

## WORTH ACTUAL MONEY

- ☐ **A smoke alarm at the charging spot**, about 10 dollars. A pack usually vents before it flames.
- ☐ **A plain steel ammo can for spare packs**, about 20 dollars. It keeps a failing spare from lighting the shelf next to it.



A pack in runaway makes its own oxygen. Powder has nothing to smother.



Spare packs in a plain steel ammo can, away from anything that burns.

### IF IT VENTS: OUT, DOOR CLOSED, PHONE

- ☐ **Do not fight it.** Fire extinguishers do not work on lithium ion battery fires, and water may not stop one from burning.
- ☐ Skip the class D lithium can. It is made for lithium metal, and there is no lithium metal in a lithium ion battery.
- ☐ No bucket, no sand. A single dump cools the outside for a moment and the cells inside keep cooking.
- ☐ Never carry a smoking pack. Burning cells can explode out of the pack.
- ☐ Get outside, stay outside, call 911. If you cannot get outside, call and tell them so.
- ☐ A pack that got cooked can fail weeks later. Treat it as live.

Watch the full video: [youtube.com/@SketchySurvival101](https://youtube.com/@SketchySurvival101)

More free sheets: [sketchysurvival101.com/free](https://sketchysurvival101.com/free)

Subscribe so the next battery fire does not catch you flat footed.